

# Difference-in-Differences with Multiple Events

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## Abstract

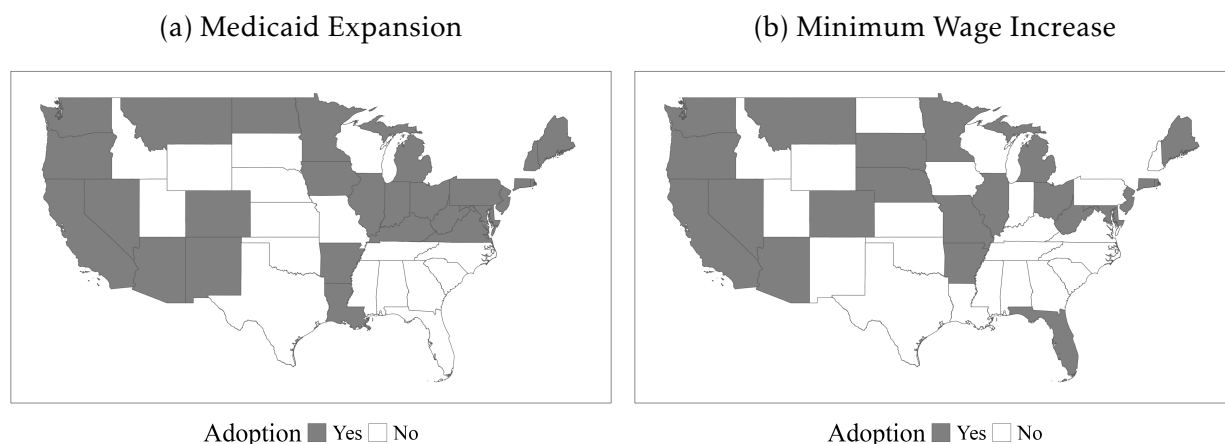
Difference-in-differences (DiD) is widely used to study the effects of state policies in the U.S., but many states adopt multiple policies around the same time. This paper studies DiD with correlated and staggered events. I first show that omitting correlated events creates an omitted event bias analogous to the omitted variable bias in cross-sectional regressions. The bias depends on the correlation between treatments and the size of the confounding treatment effect. I then propose a two-stage estimator, Double DiD, that controls for confounding events while allowing for treatment effect heterogeneity across cohorts. The first stage uses units not yet treated by any event as controls to identify the combined treatment effect. The second stage separates the target effect from confounding effects using a parallel treatment effect assumption. I apply Double DiD to study the mortality effect of Medicaid expansion, where states' minimum wage increases are positively correlated with expansion timing. The standard CS estimator finds no effect on mortality, but after controlling for minimum wage increases, Double DiD reveals a significant increase of 9.061 deaths per 100,000 in the five years after expansion. The estimator is implemented in the R package `fastdid`.

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Difference-in-Differences (DiD) is widely used to study policy impacts, especially that of the U.S. states. For example, among the citations of Callaway and Sant’Anna (2021), a popular DiD estimator, 918 papers are on the effect of minimum wages increases and 539 on the effect of Medicaid expansion from the ACA.<sup>1</sup> However, Figure 1 shows significant overlap in the states that adopt these policies. How does omitting correlated events bias DiD estimates? How can we address these bias?

Figure 1: Medicaid Expansion and Minimum Wage Increases



*Notes:* Each map shades states that adopted the policy at any point during the sample period (2009–2019). Panel (a) shades states that expanded Medicaid under the ACA. Panel (b) shades states that increased their minimum wage. States in the West, Midwest, and Northeast are more likely to have adopted both policies, while Southern states tend to have adopted neither.

This paper studies difference-in-differences with multiple staggered events. Using a setup that extends Callaway and Sant’Anna (2021) to multiple events, I first show that omitting correlated events creates a bias in DiD estimators analogous to the omitted variable bias in cross-sectional settings. I then introduce a two-stage Difference-in-Differences estimator (Double DiD) that controls for confounding events while allowing for rich treatment effect heterogeneity. The identification relies on a novel parallel treatment effect assumption and staggered adoption of confounding events.

I then apply the methods to study the mortality effect of medicaid expansion with

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<sup>1</sup>Numbers are from Google Scholar as of March 2026. The minimum wage number is from papers citing CS with exact match of “minimum wage”, the medicaid expansion number is from papers with exact match of “medicaid expansion” or “affordable care act”.

identical setup of Baker et al. (2025). From 2010 to 2020, states' minimum wage increases are positively correlated with the Medicaid Expansion under the ACA, causing the unconditional CS DiD estimator to have a bias the size of 45.50% of the effect from Medicaid Expansion. Controlling for this bias reveals a 9.061 increase in mortality per 100,000 in the 5 years after the expansion.

I contribute to the literature on confounding events by allowing for more treatment effect heterogeneity. To address the omitted event bias, 18% of the papers in *American Economic Review* with a Two-way-fixed-effects (TWFE) design control for confounding events (de Chaisemartin and D'Haultfœuille, 2020). However, after the negative weight problem became known (Goodman-Bacon, 2021), TWFE is now largely replaced by new estimators.<sup>2</sup> While the replacements are more robust to treatment effect heterogeneity, they are also less flexible in specification and cannot directly address confounding events. To the best of my knowledge, two existing methods in the modern DiD literature can account for confounding treatments: Caetano and Callaway (2024) considers time-varying covariates and de Chaisemartin and D'Haultfœuille (2023) extends de Chaisemartin and D'Haultfœuille (2020) to incorporate multiple treatments. Double DiD complements these methods by being the most flexible with the treatment effect heterogeneity of both treatments. This is achieved through being the most selective with the control group, so while Double DiD does not dominates previous contributions, it extends the frontier by providing a least-bias option in the bias-variance tradeoff.

Additionally, Fricke (2017) and Roller and Steinberg (2023) both study a two-treatment DiD in a two-period setup, Hull (2018) study a two-period setup with multiple treatment arms. Finally, Matthay et al. (2022) and Griffin et al. (2023) document the correlation among state policies and their potential confoundedness.

# 1 Identification

## 1.1 Setup

My setup extends the model of Callaway and Sant'Anna (2021) by adding a second event. Consider a panel with periods  $t \in \{1, \dots, T\}$  and units  $i \in \{1, \dots, N\} = \mathbb{I}$ .  $D_{it}^d \in \{0, 1\}$  denotes whether unit  $i$  is treated by treatment  $d \in \{1, \dots, M\} \equiv \mathbf{M}$  in period  $t$ . Without loss of

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<sup>2</sup>See Roth et al. (2022) for a review of this sprawling literature.

generality, I focus on identifying the effect from treatment 1 and call treatment 1 the target treatment and consider the rest of the treatments as confounding treatments.

Once a unit is treated, it will continue to be treated (treatment is an absorbing state):

**Assumption 1 (Irreversible Treatments)** For  $d \in \mathbf{M}$ ,  $D_{i1}^d = 0$  almost surely (a.s.), and for  $t = 2, \dots, T$   $D_{it-1}^d = 1 \implies D_{it}^d = 1$ .

With irreversible treatments, the switch to being treated can be called an event. Define the cohort of unit  $i$  for treatment  $d$  as the first period in which the event arrives,

$$G_i^d \equiv \min\{t : D_{it}^d = 1\},$$

with the convention  $G_i^d = \infty$  if the unit is never treated by  $d$ . Let  $G_{ig^d}^d = \mathbb{1}\{G_i^d = g^d\}$  denote the associated cohort indicator.

Stack the treatment-specific cohorts as  $G_i = (G_i^1, G_i^2, \dots, G_i^M)$  and write  $G_{ig} = \mathbb{1}\{G_i = \mathbf{g}\}$  for  $\mathbf{g} = (g^1, g^2, \dots, g^M)$ . I also define the never-treated cohort as  $G_{i\infty} \equiv G_{i(\infty, \infty, \dots, \infty)}$ .

For any cohort profile  $\mathbf{g}$ , let  $\min \mathbf{g} = \min_{d \in \mathbf{M}} g^d$  denote the first period in which any event occurs. Similarly, set  $\min G_i = \min_{d \in \mathbf{M}} G_i^d$ . Let  $\mathbb{G}^d$  be the support of  $G_i^d$  excluding  $\infty$  and define the collection of cohort profiles  $\mathbb{G} = \prod_{d \in \mathbf{M}} \mathbb{G}^d$ . Finally, denote the latest period with feasible untreated controls by  $\bar{t} + 1$ , where  $\bar{t} = \max_{i=1, \dots, N} \min_{d \in \mathbf{M}} G_i^d$ .

Let's generalize the dynamic potential outcome setup to  $M$  events (Murphy et al., 2001). For  $\mathbf{g} = (g^1, \dots, g^M)$  where  $g^d \in \{0, 1, \dots, T\}$  for all  $d \in \mathbf{M}$ . Let  $Y_{it}(\mathbf{g})$  denote unit  $i$ 's potential outcome in period  $t$  such that for each  $d \in D$  (1) it is first treated by event  $d$  in period  $g^d$  if  $g^d > 0$  and (2) it is never treated by event  $d$  if  $g^d = 0$ .

$$Y_{it} = Y_{it}(\mathbf{0}) + \sum_{\mathbf{g} \in \mathbb{G}} [Y_{it}(\mathbf{g}) - Y_{it}(\mathbf{0})] G_{ig}, \quad (1)$$

where  $\mathbf{0}$  is the  $M$ -dimensional vector of zeros. I assume that the observed outcome  $Y_{it}$ , treatments  $D_{it}^d$ , and covariates  $X_i \in \mathbb{R}^K$  is randomly sampled across units:

**Assumption 2 (Random Sampling)**  $\{Y_{i1}, \dots, Y_{iT}, X_i, (D_{i1}^1, \dots, D_{iT}^1), (D_{i1}^2, \dots, D_{iT}^2), \dots, (D_{i1}^M, \dots, D_{iT}^M)\}_{i=1}^N$  is independent and identically distributed (i.i.d.).

This imposes independent sampling across units but allows for arbitrary time-series autocorrelation. In the following parts, I suppress unit index  $i$ .

Finally, I impose the overlap condition:

**Assumption 3 (Overlap)** For  $d \in \mathbf{M}, t \in \{2, \dots, T\}, g^d \in \mathbb{G}$ , exists  $\epsilon > 0$  such that  $P(G_{\mathbf{g}} = 1) > \epsilon$ ,  $p_{t,\mathbf{g}}(X) < 1 - \epsilon$ , where  $p_{t,\mathbf{g}}(X) = P(G_{\mathbf{g}} = 1 | X, G_{\mathbf{g}} + (1 - D_t)(1 - G_{\mathbf{g}}) = 1)$  for  $\mathbf{g} = (g^1, g^2, \dots, g^M)$ .

This ensures that there is always some treated group and control group available and rules out irregular identification discussed in Khan and Tamer (2010).

Onto substantial assumptions:

Let  $\mathbf{x}^{(d \leftarrow z)}$  denote the vector  $\mathbf{x}$  with its  $d$ -th element set to  $z$ . My first substantial assumption is no anticipation of both treatments:

**Assumption 4 (No Anticipation)** For  $\mathbf{g} \in \mathbb{G}$ , for all  $d \in \mathbf{M}$  and  $t \in \{1, \dots, T\}$  such that  $t < g^d$ ,

$$\mathbb{E}[Y_t(\mathbf{g}) | X, G_{\mathbf{g}} = 1] = \mathbb{E}[Y_t(\mathbf{g}^{(d \leftarrow 0)}) | X, G_{\mathbf{g}} = 1].$$

No anticipation says that the events can not have a treatment effect before the event occurs. This not only holds for the periods before any event but also in periods after some events but before others. I discuss the case of no anticipation only instead of allowing limited anticipation since it adds little insight and complicates the notation greatly.

Next, the M-treatment version of the parallel trend assumption says that the untreated potential outcome untreated evolves parallelly between the treated units and units not treated by any treatments:

**Assumption 5 (Parallel Trends)** For  $\mathbf{g} \in \mathbb{G}$  and periods  $(s, t)$  such that  $t \geq \min \mathbf{g}$  and  $t \leq s \leq \bar{t}$ ,

$$\mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0}) | X, G_{\mathbf{g}} = 1] = \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0}) | X, D_s = 0].$$

where  $D_t = \max_{d \in \mathbf{M}} D_t^d$ .

Throughout this paper, I focus on the case of using not yet treated controls: the comparison group is defined by  $D_s = 0$ , so no event has occurred for those units up to period  $s$ .

The no-interaction restriction is captured by the following notation. Define  $\tau_t^{d, g^d} \equiv Y_t(\mathbf{0}^{(d \leftarrow g^d)}) - Y_t(\mathbf{0})$ .

**Assumption 6 (No Interaction)** For all  $t \in \{1, 2, \dots, T\}, \mathbf{g} \in \mathbb{G}, d \in \mathbf{M}$ ,

$$\mathbb{E}[Y_t(\mathbf{g}) - Y_t(\mathbf{g}^{(d \leftarrow 0)}) | X, G_{\mathbf{g}} = 1] = \mathbb{E}[Y_t(\mathbf{0}^{(d \leftarrow g^d)}) - Y_t(\mathbf{0}) | X, G_{\mathbf{g}} = 1].$$

## 1.2 Multiple-Group-Time Average Treatment Effect

The first stage DiD addresses the problem of confounded control units by using only units not yet treated by any event (not-yet-treated-nor-confounded) as controls. Let's first introduce the building block of the analysis, the *multiple-group-time average treatment effect on the treated* (MGTATT), defined as the average treatment effect from all treatments among the units in cohort  $G_{\mathbf{g}} = 1$  in period  $t$ :

$$ATT(\mathbf{g}, t) \equiv \mathbb{E}[Y_t(\mathbf{g}) - Y_t(\mathbf{0}) | G_{\mathbf{g}} = 1]. \quad (2)$$

I also define the target treatment effect, the parameter of interest of this paper, for  $t \in \{1, \dots, T\}$  and  $d \in \mathbf{M}$ ,  $\mathbf{g} = (g^1, g^2, \dots, g^M) \in \mathbb{G}$ ,

$$ATT^d(\mathbf{g}, t) \equiv \mathbb{E}[Y_t(\mathbf{0}^{(d \leftarrow g^d)}) - Y_t(\mathbf{0}) | G_{\mathbf{g}} = 1] \equiv \mathbb{E}[\tau_t^{1,g} | G_{\mathbf{g}} = 1], \quad (3)$$

This first stage focuses on identifying the combined effect of both treatments  $ATT(\mathbf{g}, t)$ . The discrepancy between  $ATT$  and  $ATT^1$  comes from confounded treated units and will be addressed in the second stage.

By induction and application of assumption 6 and 4, we can show that

**Lemma 1** For  $\mathbf{g} \in \mathbb{G}, t \in \{1, 2, \dots, T\}$  with  $\mathbf{g} = (g^1, g^2, \dots, g^M)$ ,

$$\mathbb{E}[Y_t(\mathbf{g}) | X, G_{\mathbf{g}} = 1] = \mathbb{E}[Y_t(\mathbf{0}) | X, G_{\mathbf{g}} = 1] + \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d,g} | X, G_{\mathbf{g}} = 1] D_t^d.$$

## 1.3 Omitted Event Bias

To understand how the DiD estimator is biased under confounding treatments, focus on the naive group-time moment with not-yet-treated controls and base period  $b \equiv g^1 - 1$ :

$$ATT^{single}(g^1, t) \equiv \mathbb{E}[Y_t - Y_b | G_{g^1}^1 = 1] - \mathbb{E}[Y_t - Y_b | D_t^1 = 0].$$

The target parameter remains

$$ATT^1(g^1, t) \equiv \mathbb{E}[\tau_t^{1,g^1} | G_{g^1}^1 = 1].$$

When there are additional events,  $ATT^{single}$  fails to recover  $ATT^1$ . The bias can be expressed compactly as follows.

**Theorem 1 (Omitted Event Bias)** Suppose there is no cohort heterogeneity so that  $ATT^d(\mathbf{g}, t) = ATT^d(t)$  for every  $d \in \mathbf{M}$  and  $\mathbf{g} \in \mathbf{G}$ . Then for any  $g^1$  and  $t \geq g^1$ ,

$$ATT^{single}(g^1, t) = ATT^1(t) + \sum_{d \in \mathbf{M} \setminus \{1\}} ATT^d(t) \Gamma_d(g^1, t),$$

where

$$\Gamma_d(g^1, t) \equiv \mathbb{E}[\mathbb{E}[D_t^d - D_b^d | X, G_{g^1}^1 = 1]] - \mathbb{E}[\mathbb{E}[D_t^d - D_b^d | X, D_t^1 = 0]].$$

The weighting term  $\Gamma_d(g^1, t)$  equals zero when exposure to event  $d$  evolves in parallel across the treated cohort and the control units, in which case the naive DiD is unbiased. Because it only depends on treatment indicators,  $\Gamma_d(g^1, t)$  can be estimated using a DiD with  $D_t^d$  as the outcome. All proofs are collected in Appendix B.

Analogous to omitted variable bias, the size of the omitted event bias is determined by the size of the confounding treatment effect ( $ATT^d$  for  $d \neq 1$ ) and the correlation between treatments:

$$\Gamma_d(g^1, t) \equiv P(D_t^d = 1, D_b^d = 0 | G_{g^1}^1 = 1) - P(D_t^d = 1, D_b^d = 0 | D_t^1 = 0), \quad (4)$$

which can be interpreted as the proportion such that  $ATT^d$  contaminates the DiD estimates. This makes  $\Gamma_d(g^1, t)$  a useful diagnostic for the potential of omitted event bias.

## 1.4 Double Difference in Differences

I again focus on the ‘not-yet’ version:

**Assumption 7 (Parallel Treatment Effects)** For  $\mathbf{g} = (g^1, g^2, \dots, g^M) \in \mathbf{G}$ ,  $\mathbf{h} \in \mathbf{G}$ ,  $t \in \{1, \dots, T\}$ ,  $d \in \mathbf{M}$ , and  $(s, t)$  such that  $t \geq g^d$ ,  $t < s$

$$ATT^d(\mathbf{g}, t) - ATT^d(\mathbf{g}, t-1) = ATT^d(\mathbf{h}^{(d \leftarrow g^d)}, t) - ATT^d(\mathbf{h}^{(d \leftarrow g^d)}, t-1),$$

This assumption sets units simultaneously treated but not yet confounded as the control group. By only using the units in the same own-cohort, I’m allowing the target treatment effect ( $ATT^1$ ) to vary freely across the treatment cohort ( $g^1$ ), and assuming it to be parallel across the confounding cohorts ( $(g^d)_{d \neq 1}$ ). For each confounding treatment effect the converse holds: free across its own cohort and parallel across the treatment cohort.

Assumption 7 plays a similar role to the common trend assumption (Assumption 4) in de Chaisemartin and D’Haultfœuille (2023). Their assumption restricts the outcome trend to be parallel conditional on the treatment status from the last period  $(D_{t-1}^1, D_{t-1}^2)$ . This means their control group is all units with the same treatment status in the last period whereas I restrict it to the units in the same own-cohort. Being more selective with control groups means that my estimator will be robust to treatment effect heterogeneity across cohorts. Caetano and Callaway (2024) focus on covariates in DiD and doesn’t explicitly discuss confounding events. Still, their method can control for confounding events by viewing them as a time-varying covariate. In such case, since their framework extends to CS’, full cohort and time heterogeneity for the target treatment is allowed. Yet how the estimator behaves under treatment effect heterogeneity of the confounding event is not discussed.

The parallel trend assumption (Assumption 5) provides a straightforward choice for the control group: units not yet treated by any event,  $D_t = 0$ . For example, with an unconditional parallel trend, we have

$$\begin{aligned}
ATT_{unc}(\mathbf{g}, t) &\equiv \mathbb{E}[Y_t - Y | G_{\mathbf{g}} = 1] - \mathbb{E}[Y_t - Y_{\min \mathbf{g} - 1} | D_t = 0] \\
&= \mathbb{E}[Y_t(\mathbf{g}) | G_{\mathbf{g}} = 1] - \mathbb{E}[Y_b(\mathbf{0}) | G_{\mathbf{g}} = 1] \\
&\quad - \mathbb{E}[Y_t(\mathbf{0}) - Y_{\min \mathbf{g} - 1}(\mathbf{0}) | D_t = 0] && \text{Assumption 4} \\
&= \mathbb{E}[Y_t(\mathbf{g}) - Y_t(\mathbf{0}) | G_{\mathbf{g}} = 1] \\
&\quad + \mathbb{E}[Y_t(\mathbf{0}) - Y_b(\mathbf{0}) | G_{\mathbf{g}} = 1] \\
&\quad - \mathbb{E}[Y_t(\mathbf{0}) - Y_{\min \mathbf{g} - 1}(\mathbf{0}) | D_t = 0] \\
&= \mathbb{E}[Y_t(\mathbf{g}) - Y_t(\mathbf{0}) | G_{\mathbf{g}} = 1] && \text{Assumption 5} \\
&= ATT(\mathbf{g}, t)
\end{aligned}$$

where  $b = \min \mathbf{g} - 1$ .

My MGTATT connects closely with the group-time average treatment effect in CS since using units not-yet-treated-nor-confounded is synonymous to using units not-yet-treated by an treatment defined as treated-or-confounded. Specifically, given a target cohort  $\mathbf{g}$ , define a subsample of the data  $\mathbb{I}(\mathbf{g})$  by removing units that are first treated by any event in  $\min \mathbf{g}$  but not in the target cohort. Formally, let, where

$$\mathbb{I}(\mathbf{g}) = \mathbb{I} \setminus \{i \in \mathbb{I} : \min G_i = \min \mathbf{g}, G_i \neq \mathbf{g}\} \quad (5)$$



In this subsample, my assumptions imply CS assumptions:

**Lemma 2** Let  $\mathbf{g} \in \mathbb{G}$  be an target cohort and  $i \in \mathbb{I}(\mathbf{g})$ , let  $\delta = 0$ ,  $g = \min \mathbf{g}$ ,  $G_{ig}^{cs} = \mathbf{1}\{\min G_i = g\}$ ,  $\mathbb{G}^{cs} = \{\min G : G \in \mathbb{G}\}$  and

$$Y_{it}^{cs}(g) = \begin{cases} Y_{it}(\mathbf{g}) & \text{if } g = \min G, \\ Y_{it}((g, g, \dots, g)) & \text{else,} \end{cases}$$

CS's Assumptions 1, 2, 3, 6 holds. Furthermore, Assumption ?? implies CS's Assumption 4, Assumption 5 implies CS's Assumption 5.

With Lemma 2, I can now introduce other identifying moments of the double-group-time average treatment effect by defining them as CS' corresponding moment in  $\mathbb{I}(\mathbf{g})$ , let

$$ATT_{ipw}(\mathbf{g}, t) \equiv \mathbb{E} \left[ \left( \frac{G_{\mathbf{g}}}{\mathbb{E}[G_{\mathbf{g}}]} - \frac{w^{comp}(\mathbf{g}, t)}{\mathbb{E}[w^{comp}(\mathbf{g}, t)]} \right) (Y_t - Y_{\min \mathbf{g}-1}) \right], \quad (6)$$

$$ATT_{or}(\mathbf{g}, t) \equiv \mathbb{E} \left[ \left( \frac{G_{\mathbf{g}}}{\mathbb{E}[G_{\mathbf{g}}]} \right) (Y_t - Y_{\min \mathbf{g}-1} - m_{\mathbf{g},t}^{ny}(X)) \right], \quad (7)$$

$$ATT_{dr}(\mathbf{g}, t) \equiv \mathbb{E} \left[ \left( \frac{G_{\mathbf{g}}}{\mathbb{E}[G_{\mathbf{g}}]} - \frac{w^{comp}(\mathbf{g}, t)}{\mathbb{E}[w^{comp}(\mathbf{g}, t)]} \right) (Y_t - Y_{\min \mathbf{g}-1} - m_{\mathbf{g},t}^{ny}(X)) \right], \quad (8)$$

with the base period  $b = \min \mathbf{g} - 1$ , the two-event version of the propensity score  $p_{\mathbf{g},t}(X) \equiv P(G_{\mathbf{g}} = 1 | X, G_{\mathbf{g}} + (1 - D_t)(1 - G_{\mathbf{g}}) = 1)$ , inverse probability weights  $w^{comp}(G_{\mathbf{g}}, t) \equiv \frac{p_{G_{\mathbf{g}},t}(X)(1 - D_t)(1 - G_{\mathbf{g}})}{1 - p_{\mathbf{g},t}(X)}$ , and outcome's control function  $m_{\mathbf{g},t}^{ny}(X) \equiv \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} | X, D_t = 0, G_{\mathbf{g}} = 0]$ . Note that these moments use none of the units excluded from  $\mathbb{I}(\mathbf{g})$ .

By showing that these moments are equivalent to CS's corresponding moments in the subsample  $\mathbb{I}(\mathbf{g})$ , Theorem 1 in CS provides identification of the double-group-time average treatment effect:

**Theorem 2** For  $\mathbf{g} \in \mathbb{G}$ ,  $t \in \{2, \dots, T\}$  such that  $\min \mathbf{g} \leq t < \bar{t}$ ,

$$ATT(\mathbf{g}, t) = ATT_{unc}(\mathbf{g}, t) = ATT_{ipw}(\mathbf{g}, t) = ATT_{or}(\mathbf{g}, t) = ATT_{dr}(\mathbf{g}, t).$$

With the confounded controls addressed, the remaining problem is the confounded treated group.

## 1.5 Second Difference in Differences

The second stage DiD addresses the problem of confounded treated units. The combined treatment effect  $ATT$  identified by the first DiD (Theorem 2) does not always equal the target  $ATT^1$  because it switches depending on the order of the events. By Assumption 4 (no anticipation), we have

$$ATT(\mathbf{g}, t) = \sum_{d \in \mathbf{M}} \mathbf{1}\{t \geq g^d\} ATT^d(\mathbf{g}, t) \quad (9)$$

In periods after event 1 but before any confounding event,  $t \geq g^1, t < g'$ , we directly observe the target parameter  $ATT^1$ . However, if the unit is also confounded ( $t \geq g'$ ), we observe instead the combined treatment effect  $ATT$ . Therefore, our goal is to impute these missing treatment effects with a new parallel trend assumption.

I have two moments for identifying the target treatment effect depending on the order of the events. First, the DiD strategy (see Figure ?? for example) is for units first confounded then treated, using units simultaneously-confounded-but-not-treated as controls.

$$ATT_{id}^{did}(\mathbf{g}, t) = ATT(\mathbf{g}, t) - ATT(\mathbf{g}, b) - \mathbb{E}[ATT(\mathbf{h}, t) - ATT(\mathbf{h}, b) | \mathbf{h} \in C^{did}(\mathbf{g}, t)] \quad (10)$$

where  $b = g^1 - 1$ ,

$$C^{did}(\mathbf{g}, t) = \{(h^1, \dots, h^M) \in \mathbb{G} : \underbrace{h^1 > t}_{\text{not yet treated}}, \underbrace{\forall d \in \mathbf{M} \text{ with } g^d \leq t, h^d = g^d}_{\text{simultaneously confounded}}\} \quad (11)$$

The second is imputation strategy (see Figure ?? for example) for units first treated then confounded, using units simultaneously-treated-but-not-yet-confounded as controls,

$$ATT_{id}^{imp}(\mathbf{g}, t) = ATT(\mathbf{g}, b) + \mathbb{E}[ATT(\mathbf{h}, t) - ATT(\mathbf{h}, b) | \mathbf{h} \in C^{imp}(\mathbf{g}, t)] \quad (12)$$

where  $b = \min_{d \in \mathbf{M} \setminus \{1\}} g^d - 1$ ,

$$C^{imp}(\mathbf{g}, t) = \{(h^1, \dots, h^M) \in \mathbb{G} : \underbrace{h^1 = g^1}_{\text{simultaneously treated}}, \underbrace{\forall d \in \mathbf{M} \setminus \{1\}, h^d > t}_{\text{not yet confounded}}\} \quad (13)$$

Combining these two strategies and the direct identification for periods before the confounding event, we can identify the target treatment effects  $ATT^1$  for cohorts with different treatment and confounding timings ( $g^1 \neq g'$ ) whenever the requisite control cohorts exist.

**Theorem 3 (Double DiD Identifies  $ATT^1$ )** *Let  $g' \equiv \min_{d \in \mathbf{M} \setminus \{1\}} g^d$  and suppose  $g^1 \neq g'$ . Under Assumptions 7 and 6, for any identifying moment  $ATT_{id}$  that equals  $ATT$ , define*

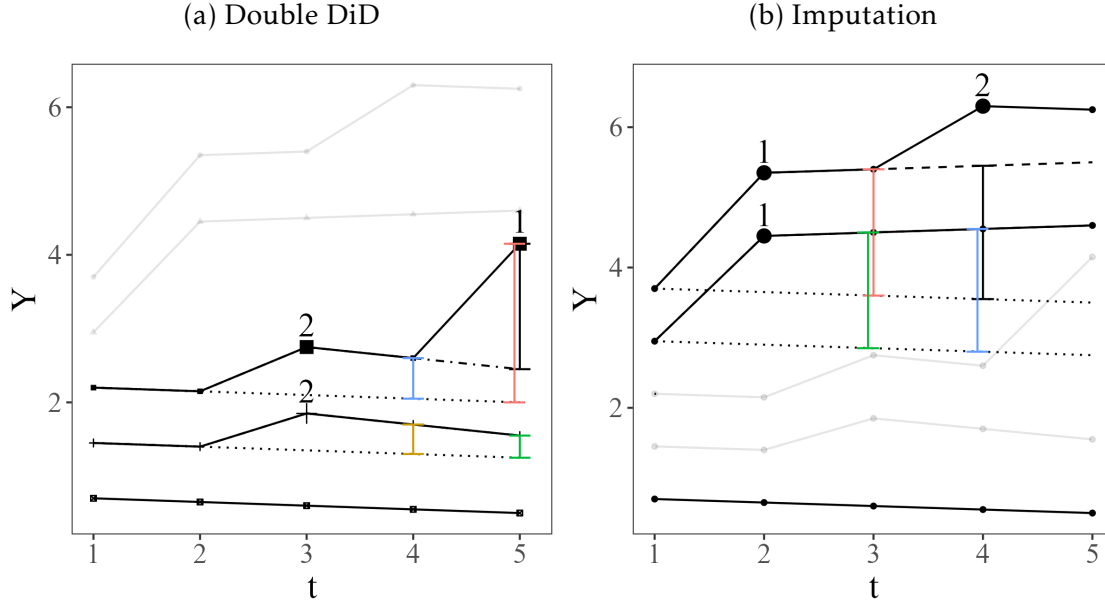
$$ATT_{id}^{dd}(\mathbf{g}, t) \equiv \begin{cases} ATT_{id}(\mathbf{g}, t), & \text{if } t < g', \\ ATT_{id}^{imp}(\mathbf{g}, t), & \text{if } t \geq g', g^1 < g', \\ ATT_{id}^{did}(\mathbf{g}, t), & \text{if } t \geq g', g^1 > g', \end{cases} \quad (14)$$

*for every period  $t$  such that the relevant control sets  $C^{imp}(\mathbf{g}, t)$  or  $C^{did}(\mathbf{g}, t)$  are non-empty. Then*

$$ATT^1(\mathbf{g}, t) = ATT_{id}^{dd}(\mathbf{g}, t).$$

Similar to the pre-trend in a single-event setup, the treatment effect in periods before event 1  $t < g^1 - 1$  is overidentified due to the parallel trend assumption. Hence the estimator can be estimated at the pre-periods to diagnose potential violation of the parallel treatment effects assumption. One limitation is that for units treated and confounded in the same period ( $g^1 = g'$ ), we cannot identify the target treatment effect due to the lack of a base period in which only one effect is observed, similar to the units treated in the first period in the standard staggered setup.

Figure 2: Identification



This figure illustrates the  $M = 2$  case. **Left:** Three cohorts are shown: a treated group with  $\mathbf{g} = (5, 3)$ , a control group for imputing the confounding treatment effect with  $\mathbf{g} = (\infty, 3)$ , and a never-treated group  $\mathbf{g} = (\infty, \infty)$ . **Right:** Three cohorts: a treated group with  $\mathbf{g} = (2, 4)$ , a control group for imputing the target treatment effect with  $\mathbf{g} = (2, \infty)$ , and a never-treated group  $\mathbf{g} = (\infty, \infty)$ . After identifying the DGTATs (colorful bars) with Theorem 2, the double DiD design compares the treatment effect trend of the control group (green bar – yellow bar) and the treated group (red bar – blue bar). The difference is the target treatment effect (black bar) under the parallel treatment effect assumption.

## 2 Aggregation, Estimation, and Inference

Given the connection in identification, the aggregation, estimation, and inference results are also similar to CS. To aggregate the double-group-time target treatment effects from the second stage, each summary parameter  $\theta$  is associated with a weighting scheme,  $w(\mathbf{g}, t)$  such that

$$\theta = \sum_{\mathbf{g} \in \mathbb{G}} \sum_{t=2}^T w(\mathbf{g}, t) \text{ATT}^1(\mathbf{g}, t). \quad (15)$$

For example, to obtain the group-time average treatment effect,  $\text{ATT}^1(\dot{\mathbf{g}}, \dot{t})$ , there is the weighting scheme

$$w_{\dot{g},t}(\mathbf{g}, t) = \mathbb{1}\{g^1 = \dot{g}\} \mathbb{1}\{t = \dot{t}\} A(\mathbf{g}, t) P(G_{\mathbf{g}} = 1 | G_{\dot{g}}^1 = 1, A(\mathbf{g}, t) = 1), \quad (16)$$

where  $A(\mathbf{g}, t) \equiv \mathbb{1}\{g^1 < g'\} \mathbb{1}\{g^1 \leq t, C^{imp}(\mathbf{g}, t) \neq \emptyset\} + \mathbb{1}\{g^1 > g'\} \mathbb{1}\{g^1 \leq t, C^{did}(\mathbf{g}, t) \neq \emptyset\}$  is the indicator for availability of a suitable control group, with  $g' = \min_{d \in \mathbf{M} \setminus \{1\}} g^d$ . With the group-time parameter, the other aggregated parameters can be done using the summary scheme in CS.

For estimation, the first stage moments from Theorem 2 can be estimated with sample plug-ins. Following CS, I focus on the doubly-robust version since it has the benefit of additional robustness to misspecification and the arguments apply analogously to other parameters. For a generic  $C \in \{0, 1\}, Z$ , let  $\mathbb{E}_n[Z|C]$  denote the conditional summation  $(\sum_{i=1}^N C_i)^{-1} \sum_{i=1}^N C_i Z_i$ . Let

$$\widehat{ATT}_{dr}(\mathbf{g}, t) \equiv \mathbb{E}_n \left[ \left( \hat{w}_{\mathbf{g}}^{treat} - \hat{w}_{\mathbf{g}}^{comp} \right) (Y_t - Y_{\min_{\mathbf{g}-1}} - \hat{m}_{\mathbf{g},t}(X)) | i \in \mathbb{I}(\mathbf{g}) \right] \quad (17)$$

where

$$\hat{w}_{\mathbf{g}}^{treat} = \frac{G_{\mathbf{g}}}{\mathbb{E}_n[G_{\mathbf{g}} | i \in \mathbb{I}(\mathbf{g})]}, \quad \hat{w}_{\mathbf{g}}^{comp,ny} = \frac{\frac{\hat{p}_{\mathbf{g},t}(X)(1-D_t)(1-G_{\mathbf{g}})}{1-\hat{p}_{\mathbf{g},t}(X)(1-D_t)(1-G_{\mathbf{g}})}}{\mathbb{E}_n \left[ \frac{\hat{p}_{\mathbf{g},t}(X)(1-D_t)(1-G_{\mathbf{g}})}{1-\hat{p}_{\mathbf{g},t}(X)(1-D_t)(1-G_{\mathbf{g}})} | i \in \mathbb{I}(\mathbf{g}) \right]},$$

and  $\hat{p}(\cdot), \hat{m}(\cdot)$  is some estimators of  $p(\cdot), m(\cdot)$  as defined for Equation 8.

For the second stage moments in Theorem 3, the formulation in Equation 18 and 19 means that we can treat them as a summary parameter of the first stage estimates and aggregate them as weighted summations similar to Equation 15.

$$ATT_{id}^{did}(\tilde{\mathbf{g}}, \tilde{t}) = \sum_{\mathbf{g} \in \mathbb{G}} \sum_{t=2}^T w_{\tilde{\mathbf{g}}, \tilde{t}}^{did}(\mathbf{g}, t) ATT_{id}(\mathbf{g}, t), \quad (18)$$

$$w_{\tilde{\mathbf{g}}, \tilde{t}}^{did}(\mathbf{g}, t) \equiv \begin{cases} 1, & \text{if } t = \tilde{t}, \mathbf{g} = \tilde{\mathbf{g}}, \\ -1, & \text{if } t = \tilde{g}^1 - 1, \mathbf{g} = \tilde{\mathbf{g}}, \\ -P(G_{\mathbf{g}} = 1 | \mathbf{g} \in C^{did}(\tilde{\mathbf{g}}, \tilde{t})), & \text{if } \mathbf{g} \in C^{did}(\tilde{\mathbf{g}}, \tilde{t}), t = \tilde{t}, \\ P(G_{\mathbf{g}} = 1 | \mathbf{g} \in C^{did}(\tilde{\mathbf{g}}, \tilde{t})), & \text{if } \mathbf{g} \in C^{did}(\tilde{\mathbf{g}}, \tilde{t}), t = \tilde{g}^1 - 1, \\ 0, & \text{else.} \end{cases}$$

$$ATT_{id}^{imp}(\tilde{\mathbf{g}}, \tilde{t}) = \sum_{\mathbf{g} \in \mathbb{G}} \sum_{t=2}^T w_{\tilde{\mathbf{g}}, \tilde{t}}^{imp}(\mathbf{g}, t) ATT_{id}(\mathbf{g}, t), \quad (19)$$

$$w_{\tilde{\mathbf{g}}, \tilde{t}}^{imp}(\mathbf{g}, t) \equiv \begin{cases} 1, & \text{if } t = \tilde{g}' - 1, \mathbf{g} = \tilde{\mathbf{g}}, \\ P(G_{\mathbf{g}} = 1 \mid \mathbf{g} \in C^{imp}(\tilde{\mathbf{g}}, \tilde{t})), & \text{if } \mathbf{g} \in C^{imp}(\tilde{\mathbf{g}}, \tilde{t}), t = \tilde{t}, \\ -P(G_{\mathbf{g}} = 1 \mid \mathbf{g} \in C^{imp}(\tilde{\mathbf{g}}, \tilde{t})), & \text{if } \mathbf{g} \in C^{imp}(\tilde{\mathbf{g}}, \tilde{t}), t = \tilde{g}' - 1, \\ 0, & \text{else.} \end{cases}$$

Since the double-group-time average treatment effect estimators are equivalent to CS' estimators for group-time average treatment effects in  $\mathbb{I}(\mathbf{g})$  (for example, Equation 17 corresponds to Equation 4.2 in CS), and the double DiD estimators and other summary estimators are their aggregations CS' inference procedure in Section 4.1 applies. In summary, by assuming that the control function and propensity score can be estimated by smooth parametric models with  $\sqrt{N}$  linear representation (CS Assumption 7) and some weak integrability conditions that ensure the second moments are finite (CS Assumption 8), CS Theorem 2 provides the influence function for calculating the standard error and its asymptotic normality. Then, CS Theorem 3 establishes the validity of the multiplier bootstrap for calculating the standard errors. The estimator is implemented in the R package *fastdid*.

### 3 Empirical Application: Medicaid Expansion, Minimum Wages, and Mortality

To illustrate the empirical relevance of omitted event bias, I study the mortality effect of Medicaid expansion following an identical setup to Baker et al. (2025). I define treatment cohorts by the timing of Medicaid expansion under the ACA and use counties in states that never expanded Medicaid as control groups (never-treated). I choose minimum wage increases as the confounding event. While discussion around minimum wage focused mainly on its labor market impact (Dube, Lester and Reich, 2016; Cengiz et al., 2019), a number of works have also documented its effect on health and mortality.<sup>3</sup> I define a state's minimum wage hike cohort as the year in which the state first increases minimum wage (The federal minimum wage remained fixed at \$7.25 per hour since 2009).

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<sup>3</sup>Dow et al. (2020) find significant effect of minimum wage on suicides, other potential mechanism include crowding out employer health insurance (Meiselbach and Abraham, 2023), contribution to social security (Kim, Lee and Lee, 2023), and driving under influence (Adams, Blackburn and Cotti, 2012). See Neumark (2024) and Paul Leigh, Leigh and Du (2019) for a comprehensive review of the effect of minimum wages.

I construct a county-year level dataset that spans 2009 to 2019 using county mortality data from Baker et al. (2025), supplemented with state minimum wage history from Sorens, Muedini and Ruger (2008) and state characteristics from University of Kentucky Center for Poverty Research (2024). The outcome is the crude mortality rate per 100,000 population aged 20–64. DC, Delaware, Massachusetts, New York, and Vermont are excluded as states that expanded Medicaid before the study period. Table 1 reports summary statistics by Medicaid expansion status. Counties in Medicaid-expanding states are also more likely to be in states that increased their minimum wages and to have a democrat governor.

Table 1: State Characteristics

|                    | Non-Adopt | Adopt  | Other  |
|--------------------|-----------|--------|--------|
| White (%)          | 77.90     | 79.50  | 79.70  |
| Female (%)         | 50.50     | 50.10  | 50.30  |
| Median income      | 49,313    | 57,861 | 55,340 |
| Unemployment (%)   | 6.995     | 8.011  | 6.742  |
| Medicaid Rate      | 0.160     | 0.205  | 0.166  |
| Democrat Gov.      | 0.055     | 0.674  | 0.026  |
| Min. wage          | 7.350     | 7.850  | 7.284  |
| Min. wage increase | 0.235     | 0.773  | 0.429  |
| County count       | 1,222     | 978    | 404    |
| State count        | 17        | 22     | 7      |

Note: The table reports weighted means of county characteristics in 2013 by Medicaid expansion status. Non-Adopt: states that never expanded Medicaid. Adopt: states that expanded in 2014. Other: states that expanded after 2014. Counties are weighted by the 2013 population aged 20–64.

Figure 3 reports the mortality trend by Medicaid expansion status, split by whether states increased their minimum wage. Panel (a) shows that in the pooled sample, mortality trends are roughly parallel between adopting and non-adopting states, consistent with the null CS estimate. However, Panels (c) reveal that among states that did not in-

crease their minimum wage (Panel c), adopting states show a clear upward trend relative to non-adopters after expansion.

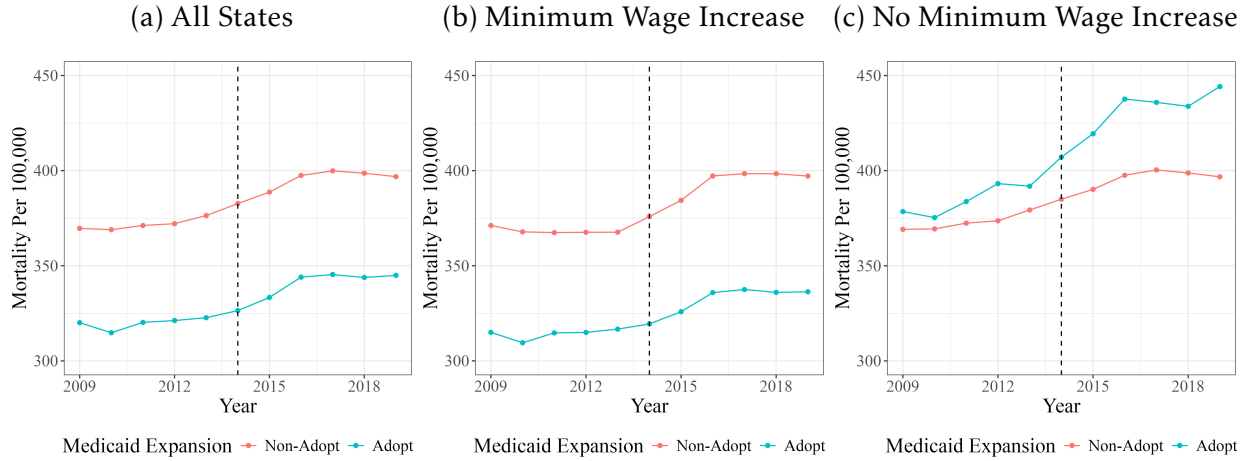
Figure 4(a) reports the correlation between Medicaid expansion and minimum wage increases using the diagnostic in Equation 4. States that expanded Medicaid also tend to increase their minimum wages shortly after. For example, in a staggered DiD design with unconditional estimators, the treatment effect estimates will have an average bias with the size of 45.50% of the Medicaid expansion effect. Alarming, the correlation was not as significant before the Medicaid expansion, suggesting that pre-trends would not have been a good diagnostic in this case.

Figure 4(b)–(c) report the effects of Medicaid expansion on mortality per 100,000. With a staggered DiD with CS’ unconditional estimator, I find a no effect on mortality, consistent with Baker et al. (2025). However, when I use the Double DiD estimator to control for minimum wage increases, we find a significant effect of 9.061 per 100,000 in the 5 years after the expansion.

The discrepancy between the CS and double DiD estimates of Medicaid expansion’s mortality effect does not necessarily contradict previous empirical findings. The empirical literature often controls for multiple confounding policies and uses rich covariate specifications, which is likely to weaken the omitted event bias from minimum wage increases. Additionally, other correlated state policies may further confound both estimates. The takeaway is that in an important application of the DiD design, the omitted event bias from correlated state policies is empirically relevant.

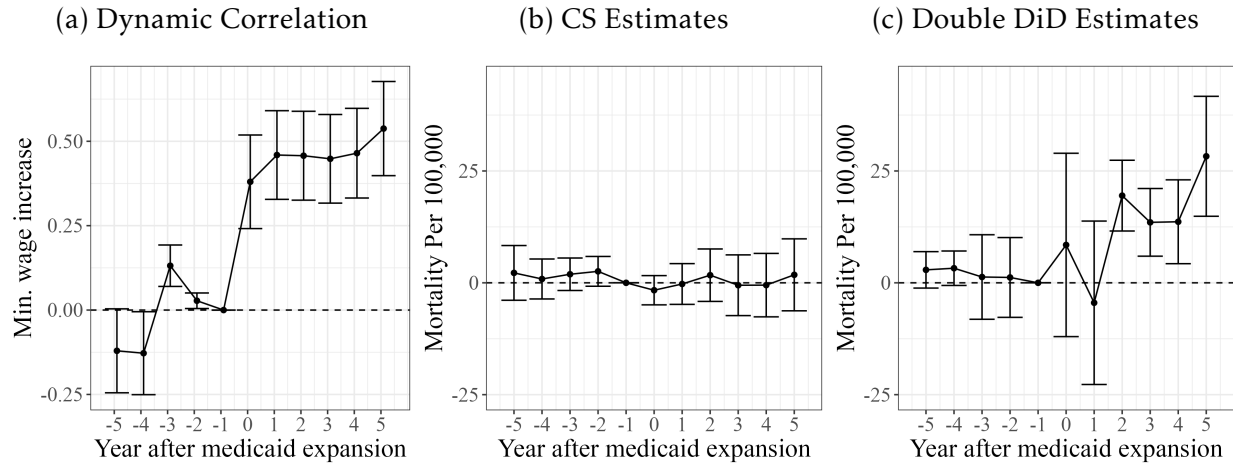


Figure 3: Mortality trend by medicaid expansion and minimum wage increases



Panel (a) reports mortality trend among all states by medicaid expansion adoption (states in the “other” group are excluded). Panel (b) reports the trend among states that increased minimum wage between 2010 and 2019. Panel (c) reports the trend among states that did not increased minimum wage.

Figure 4: The effect of medicaid expansion controlling for minimum wage increases



Panel (a) reports the event study estimate of the probability of minimum wage increase for Medicaid expansion cohorts, using the diagnostic in Equation 4. Panels (b) and (c) report event study estimates of the effect of Medicaid expansion on mortality per 100,000 population aged 20–64. Panel (b) uses the unconditional CS estimator and panel (c) uses the Double DiD estimator. The x-axis is years relative to Medicaid expansion. Points and error bars report the estimated average treatment effect and 95% confidence interval. Standard errors are clustered at the county level.

## 4 Discussion

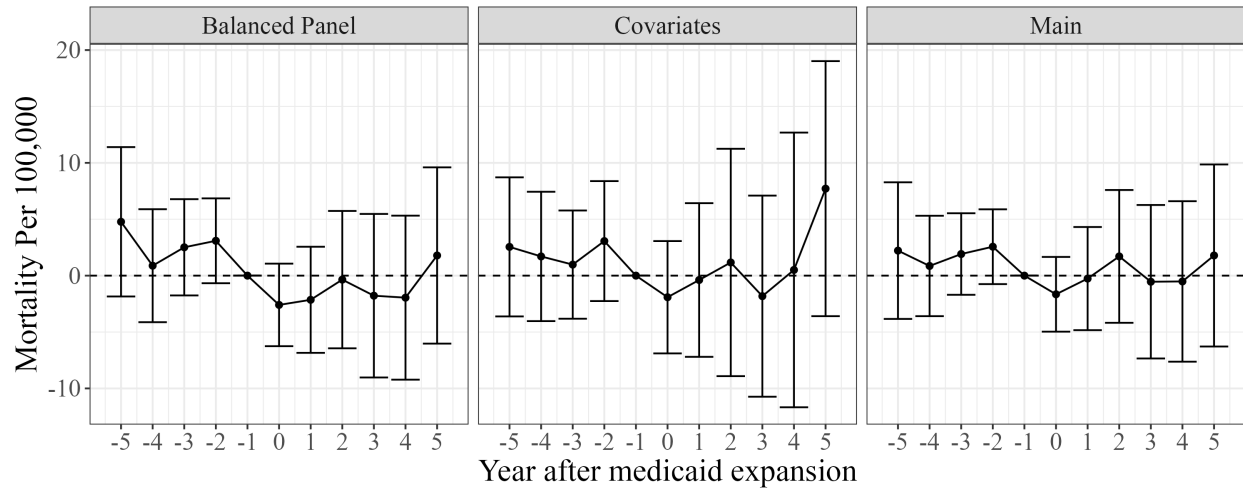
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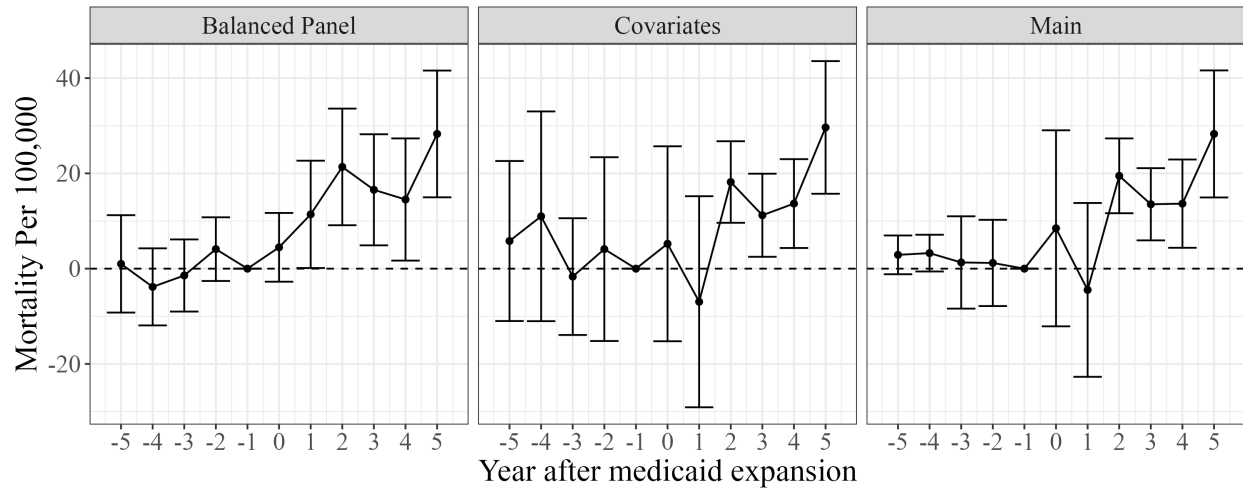
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## **Appendix A   Additional Results**



(a) CS Estimates



(b) Double DiD Estimates

Figure 5: Estimates of Medicaid Expansion on Mortality under Alternative Specifications

This figure reports event study estimates of the effect of Medicaid expansion on mortality per 100,000 population aged 20–64 across alternative specifications, corresponding to the main specification in Figure 4. “Covariates” adds demographic controls (race, sex, Hispanic ethnicity, unemployment, poverty, and median income) as in Baker et al. (2025). “Balanced Panel” restricts to a balanced panel of event times.

## Appendix B Proofs

### B.1 Proof of Theorem 1

Within this proof, I denote the base period by  $b \equiv g - 1$ . Let's first decompose the moment into four parts

Let  $\mathbb{G}_g = \{(g^1, g^2, \dots, g^M) \in \mathbb{G} : g^1 = g\}$  denote the cohorts first treated by the first event in  $g$ . For the treated group in  $t$ ,

$$\begin{aligned}
 \mathbb{E}[Y_t|X, G_g^1 = 1] &= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_t|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_t(\mathbf{g})|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \mathbb{E}[Y_t(\mathbf{0})|X, G_g^1 = 1] + \sum_{\mathbf{g} \in \mathbb{G}_g} \left[ \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d, g^d}|X, G_{\mathbf{g}} = 1]D_t^d \right] P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \mathbb{E}[Y_t(\mathbf{0})|X, G_g^1 = 1] + \sum_{d \in \mathbf{M}} \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[\tau_t^{d, g^d}|X, G_{\mathbf{g}} = 1]D_t^d P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \mathbb{E}[Y_t(\mathbf{0})|X, G_g^1 = 1] + \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d, g^d}|X, D_t^d = 1, G_g^1 = 1]P(D_t^d = 1|X, G_g^1 = 1)
 \end{aligned}$$

For the treated group in  $b$ ,

$$\begin{aligned}
 \mathbb{E}[Y_b|X, G_g^1 = 1] &= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_b|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_b(\mathbf{g})|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \mathbb{E}[Y_b(\mathbf{0})|X, G_g^1 = 1] + \sum_{\mathbf{g} \in \mathbb{G}_g} \left[ \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_b^{d, g^d}|X, G_{\mathbf{g}} = 1]D_b^d \right] P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \mathbb{E}[Y_b(\mathbf{0})|X, G_g^1 = 1] + \sum_{d \in \mathbf{M}} \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[\tau_b^{d, g^d}|X, G_{\mathbf{g}} = 1]D_b^d P(G_{\mathbf{g}} = 1|X, G_g^1 = 1) \\
 &= \mathbb{E}[Y_b(\mathbf{0})|X, G_g^1 = 1] + \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_b^{d, g^d}|X, D_b^d = 1, G_g^1 = 1]P(D_b^d = 1|X, G_g^1 = 1)
 \end{aligned}$$



For the control group in  $t$

$$\begin{aligned}
\mathbb{E}[Y_t|X, D_t^1 = 0] &= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_t|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_t(\mathbf{g})|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \mathbb{E}[Y_t(\mathbf{0})|X, D_t^1 = 0] + \sum_{\mathbf{g} \in \mathbb{G}_g} \left[ \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d, g^d} | X, G_{\mathbf{g}} = 1] D_t^d \right] P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \mathbb{E}[Y_t(\mathbf{0})|X, D_t^1 = 0] + \sum_{d \in \mathbf{M}} \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[\tau_t^{d, g^d} | X, G_{\mathbf{g}} = 1] D_t^d P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \mathbb{E}[Y_t(\mathbf{0})|X, D_t^1 = 0] + \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d, g^d} | X, D_t^d = 1, D_t^1 = 0] P(D_t^d = 1|X, D_t^1 = 0)
\end{aligned}$$

For the control group in  $b$

$$\begin{aligned}
\mathbb{E}[Y_b|X, D_t^1 = 0] &= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_b|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[Y_b(\mathbf{g})|X, G_{\mathbf{g}} = 1]P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \mathbb{E}[Y_b(\mathbf{0})|X, D_t^1 = 0] + \sum_{\mathbf{g} \in \mathbb{G}_g} \left[ \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_b^{d, g^d} | X, G_{\mathbf{g}} = 1] D_b^d \right] P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \mathbb{E}[Y_b(\mathbf{0})|X, D_t^1 = 0] + \sum_{d \in \mathbf{M}} \sum_{\mathbf{g} \in \mathbb{G}_g} \mathbb{E}[\tau_b^{d, g^d} | X, G_{\mathbf{g}} = 1] D_b^d P(G_{\mathbf{g}} = 1|X, D_t^1 = 0) \\
&= \mathbb{E}[Y_b(\mathbf{0})|X, D_t^1 = 0] + \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_b^{d, g^d} | X, D_b^d = 1, D_t^1 = 0] P(D_b^d = 1|X, D_t^1 = 0)
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathbb{E}[Y_t - Y_b|X, G_g^1 = 1] - \mathbb{E}[Y_t - Y_b|X, D_t^1 = 0] &= \mathbb{E}[Y_t|X, G_g^1 = 1] - \mathbb{E}[Y_b|X, G_g^1 = 1] - \mathbb{E}[Y_t|X, D_t^1 = 0] + \mathbb{E}[Y_b|X, D_t^1 = 0] \\
&= \mathbb{E}[Y_t(\mathbf{0})|X, G_g^1 = 1] - \mathbb{E}[Y_b(\mathbf{0})|X, G_g^1 = 1] + \mathbb{E}[Y_t(\mathbf{0})|X, D_t^1 = 0] - \mathbb{E}[Y_b(\mathbf{0})|X, D_t^1 = 0] \\
&\quad + \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d, g^d} | X, D_t^d = 1, G_g^1 = 1] P(D_t^d = 1 | X, G_g^1 = 1) \\
&\quad - \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_b^{d, g^d} | X, D_b^d = 1, G_g^1 = 1] P(D_b^d = 1 | X, G_g^1 = 1) \\
&\quad - \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d, g^d} | X, D_t^d = 1, D_t^1 = 0] P(D_t^d = 1 | X, D_t^1 = 0) \\
&\quad + \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_b^{d, g^d} | X, D_b^d = 1, D_t^1 = 0] P(D_b^d = 1 | X, D_t^1 = 0) \\
&= \sum_{d \in \mathbf{M}} \mathbb{E}[\tau_t^{d, g^d} | X, D_t^d = 1, G_g^1 = 1] \left[ P(D_t^d = 1, D_b^d = 0 | X, G_g^1 = 1) - P(D_t^d = 1, D_b^d = 1 | X, G_g^1 = 1) \right] \\
&\quad - \sum_{d \in \mathbf{M}} \left[ \mathbb{E}[\tau_b^{d, g^d} | X, D_b^d = 1, G_g^1 = 1] - \mathbb{E}[\tau_t^{d, g^d} | X, D_t^d = 1, G_g^1 = 1] \right] P(D_t^d = 1, D_b^d = 1 | X, G_g^1 = 1) \\
&\quad - \sum_{d \in \mathbf{M}} \left[ \mathbb{E}[\tau_t^{d, g^d} | X, D_t^d = 1, D_t^1 = 0] - \mathbb{E}[\tau_b^{d, g^d} | X, D_b^d = 1, D_t^1 = 0] \right] P(D_t^d = 1, D_b^d = 1 | X, D_t^1 = 0) \\
&\quad + \sum_{d \in \mathbf{M}} \left[ \mathbb{E}[\tau_b^{d, g^d} | X, D_b^d = 1, D_t^1 = 0] - \mathbb{E}[\tau_t^{d, g^d} | X, D_t^d = 1, D_t^1 = 0] \right] P(D_t^d = 1, D_b^d = 1 | X, D_t^1 = 0)
\end{aligned}$$

Imposing  $ATT^d(\mathbf{g}, s) = ATT^d(s)$  for all  $d$  and all  $s$  implies that the conditional expectations of  $\tau_s^{d, g^d}$  in the display above equal  $ATT^d(s)$  whenever the conditioning event has  $D_s^d = 1$  and equal zero when  $D_s^d = 0$  by Assumption 4. Re-arranging terms and integrating both sides with respect to  $X$  therefore yields

$$ATT^{single}(g, t) = ATT^1(t) + \sum_{d \in \mathbf{M} \setminus \{1\}} ATT^d(t) \Gamma_d(g, t),$$

which proves Theorem 1.

## B.2 Proof of Lemma 2

In the following parts, I abbreviate CS's assumption X as CSX and my assumption X as AX.

Note that for all people,  $Y_t^{cs}(0) = Y_t(\mathbf{0})$  since if  $\min G = 0$ ,  $G = \mathbf{0}$  and  $Y_t^{cs}(0) = Y_t(\mathbf{0})$ , else  $Y_t^{cs}(0) = Y_t((0, \dots, 0)) = Y_t(\mathbf{0})$ .

First, notice that  $Y_t((g, g, \dots, g)) = Y_t^{cs}(g)$ : if  $G_i = (g, \dots, g)$  then  $\min G_i = g$  and  $Y_t^{cs}(g) = Y_{it}(\mathbf{g}) = Y_{it}(g, \dots, g)$ ; otherwise  $Y_t^{cs}(g) = Y_{it}(g, \dots, g)$  by definition. Also  $D_t = \max_{d \in \mathbf{M}} D_t^d$  since  $D_t \equiv \mathbb{1}\{\min_{d \in \mathbf{M}} G_i^d \leq t\} = \max_{d \in \mathbf{M}} \mathbb{1}\{G_i^d \leq t\} = \max_{d \in \mathbf{M}} D_t^d$ .

Always with  $\delta = 0$ .

**Claim 1 (CS1, Irreversibility of Treatment)**  $D_1 = 0$  almost surely. For  $t = 2, \dots, T$ ,  $D_{t-1} = 1$  implies  $D_t = 1$ .

Proof: By A1,  $D_1^d = 0$  a.s. for all  $d \in \mathbf{M}$ , so  $D_1 = \max_{d \in \mathbf{M}} D_1^d = 0$  a.s. For the second part, take  $t = 2, \dots, T$ : if  $D_{t-1} = 1$ , then there exists  $d \in \mathbf{M}$  such that  $D_{t-1}^d = 1$ , so by A1,  $D_t^d = 1$  and thus  $D_t = 1$ .

**Claim 2 (CS2.1, Potential Outcome)**  $Y_{it} = Y_{it}^{cs}(0) + \sum_{g=2}^T (Y_{it}^{cs}(g) - Y_{it}^{cs}(0)) G_{ig}^{cs}$ .

Proof:

$$\begin{aligned}
& Y_{it}^{cs}(0) + \sum_{g=2}^T (Y_{it}^{cs}(g) - Y_{it}^{cs}(0)) G_{ig}^{cs} \\
&= Y_{it}^{cs}(0) + (Y_{it}^{cs}(\min G_i) - Y_{it}^{cs}(0)) & G_{ig}^{cs} = \mathbf{1}\{\min G_i = g\} \\
&= Y_{it}^{cs}(0) + (Y_{it}(G) - Y_{it}^{cs}(0)) & Y_{it}^{cs}(g) = Y_{it}(\mathbf{g}) \text{ if } \min G = g, \\
&= Y_{it}(\mathbf{0}) + (Y_{it}(G) - Y_{it}(\mathbf{0})) & \text{if } \min G = 0, \text{ then } G = \mathbf{0}, \text{ o.w., } Y_{it}^{cs}(0) = Y_{it}(\mathbf{0}) \\
&= Y_{it}(\mathbf{0}) + \sum_{\mathbf{g} \in \mathbb{G}} (Y_{it}(\mathbf{g}) - Y_{it}(\mathbf{0})) G_{\mathbf{g}} & G_{\mathbf{g}} = 0 \text{ if } G \neq \mathbf{g} \\
&= Y_{it} & \text{Equation 1.}
\end{aligned}$$

**Claim 3 (CS2, Random Sampling)**  $\{Y_{i1}, Y_{i2}, \dots, Y_{iT}, X_i, D_{iT}, D_{iT}, \dots, D_{iT}\}$  are i.i.d. across  $i$ .

Proof:  $Y, X$  are the same since we only subsample by units.  $D_t^d$  is i.i.d. (A2) for all  $d \in \mathbf{M}$ , which implies  $D_t = \max_{d \in \mathbf{M}} D_t^d$  is i.i.d.

**Claim 4 (CS3, Limited Treatment Anticipation)**  $\mathbb{E}[Y^{cs}(g)|X, G_g^{cs} = 1] = \mathbb{E}[Y^{cs}(0)|X, G_g^{cs}] = 1$  a.s. for all  $g \in \mathbb{G}^{cs}, t \in \{1, \dots, T\}$ . such that  $t < g$ .

Proof:

$$\begin{aligned}
& E[Y_t^{cs}(g)|X, G_g^{cs} = 1] \\
&= E[Y_t((g, g, \dots, g))|X, G_g^{cs} = 1, \min G \neq g](1 - P(\min G = g|X, G_g^{cs} = 1)) \\
&\quad + E[Y_t(G)|X, G_g^{cs} = 1, \min G = g]P(\min G = g|X, G_g^{cs} = 1) \\
&= E[Y_t(\mathbf{0})|X, G_g^{cs} = 1, \min G \neq g](1 - P(\min G = g|X, G_g^{cs} = 1)) \quad g > t, A4 \\
&\quad + E[Y_t(G)|X, G_g^{cs} = 1, \min G = g]P(\min G = g|X, G_g^{cs} = 1) \\
&= E[Y_t(\mathbf{0})|X, G_g^{cs} = 1, \min G \neq g](1 - P(\min G = g|X, G_g^{cs} = 1)) \quad \forall d \in \mathbf{M}, g^d \geq \min G = g > t, A4 \\
&\quad + E[Y_t(\mathbf{0})|X, G_g^{cs} = 1, \min G = g]P(\min G = g|X, G_g^{cs} = 1) \\
&= E[Y_t(\mathbf{0})|X, G_g^{cs} = 1] \\
&= E[Y_t(0)|X, G_g^{cs} = 1].
\end{aligned}$$

**Claim 5 (CS4, Conditional Parallel Trends Based on a “Never-Treated” Group)**  $E[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, G_g^{cs} = 1] = E[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, G_\infty^{cs} = 1]$  a.s. for all  $g \in \mathbb{G}^{cs}, t \geq g$ .

Proof: take  $t \geq g$ ,

$$\begin{aligned}
& \mathbb{E}[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, G_g^{cs} = 1] \\
&= \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, G_g^{cs} = 1] \\
&= \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, \min G = g] \quad G_g^{cs} \equiv \min G \\
&= \sum_{\mathbf{g} \in \{\mathbf{g} \in \mathbb{G} : \min \mathbf{g} = g\}} P(G_{\mathbf{g}} = 1|X, \min G = g) \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, G_{\mathbf{g}} = 1] \\
&= \sum_{\mathbf{g} \in \{\mathbf{g} \in \mathbb{G} : \min \mathbf{g} = g\}} P(G_{\mathbf{g}} = 1|X, \min G = g) \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, G_\infty = 1] \quad t \geq g = \min \mathbf{g}, A?? \\
&= \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, G_\infty = 1] \\
&= \mathbb{E}[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, G_\infty = 1] \\
&= \mathbb{E}[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, G_\infty^{cs} = 1].
\end{aligned}$$

**Claim 6 (CS5, Conditional Parallel Trends Based on a “Not-Yet-Treated” Group)**  $E[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, G_g^{cs} = 1] = E[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, D_t^{cs} = 0, G_g^{cs} = 0]$  a.s. for all  $t \geq g, t \leq s < \bar{t}^{cs}$ .

Proof: take  $t \geq g, t \leq s < \bar{t}$

$$\begin{aligned}
& \mathbb{E}[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, G_g^{cs} = 1] \\
&= \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, G_g^{cs} = 1] \\
&= \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, \min G = g] \quad G_g^{cs} \equiv \min G \\
&= \sum_{\mathbf{g} \in \{\mathbf{g} \in \mathbb{G}: \min \mathbf{g} = g\}} P(G_{\mathbf{g}} = 1|X, \min G = g) \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, G_{\mathbf{g}} = 1] \\
&= \sum_{\mathbf{g} \in \{\mathbf{g} \in \mathbb{G}: \min \mathbf{g} = g\}} P(G_{\mathbf{g}} = 1|X, \min G = g) \mathbb{E}[Y_t(\mathbf{0}) - Y_{t-1}(\mathbf{0})|X, D_s = 0, G_{\mathbf{g}} = 0] \quad t \geq g = \min \mathbf{g}, t \leq s < \bar{t}, A5 \\
&= \sum_{\mathbf{g} \in \{\mathbf{g} \in \mathbb{G}: \min \mathbf{g} = g\}} P(G_{\mathbf{g}} = 1|X, \min G = g) \mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, D_s = 0, G_{\mathbf{g}} = 0] \\
&= \mathbb{E}[Y_t^{cs}(0) - Y_{t-1}^{cs}(0)|X, D_s = 0, G_g^{cs} = 0]
\end{aligned}$$

**Claim 7 (CS6, Overlap)** For each  $t \in \{2, \dots, T\}$ ,  $g \in \mathbb{G}^{cs}$ , there exists some  $\epsilon > 0$  such that  $P(G_g^{cs} = 1) > \epsilon$  and  $p_{g,t}^{cs}(X) < 1 - \epsilon$  almost surely, where  $p_{g,t}^{CS}(X) \equiv P(G_g^{cs} = 1|X, G_g^{cs} + (1 - D_t)(1 - G_g^{cs}) = 1)$ .

Proof: take  $g \in \mathbb{G}^{cs}$ , by construction there exists  $\mathbf{g} \in \mathbb{G}$  such that  $g = \mathbf{g}$ ,

$$P(G_g^{cs} = 1) = \sum_{\mathbf{g} \in \mathbb{G}, \min \mathbf{g} = g} P(G_{\mathbf{g}} = 1) > \sum_{\mathbf{g} \in \mathbb{G}, \min \mathbf{g} = g} \epsilon_{\mathbf{g}} > 0 \quad A3,$$

where  $\epsilon_{\mathbf{g}}$  is the  $\epsilon$  from Assumption 3,  $\epsilon_{\mathbf{g}} > 0$  for all  $\mathbf{g} \in \mathbb{G}$ . For the second part,

$$\begin{aligned}
p_{g,t}^{cs}(X) &= P(G_g^{cs} = 1|X, G_g^{cs} + (1 - D_t)(1 - G_g^{cs}) = 1) \\
&= \frac{P(G_g^{cs} = 1|X)}{P(G_g^{cs} + (1 - D_t)(1 - G_g^{cs}) = 1|X)} \\
&= \frac{P(G_{\mathbf{g}} = 1|X) + \sum_{\mathbf{h} \in \mathbb{G} \setminus \{\mathbf{g}\}} P(G_{\mathbf{h}} = 1|X)}{P(G_{\mathbf{g}} + (1 - D_t)(1 - G_{\mathbf{g}}) = 1|X) + \sum_{\mathbf{h} \in \mathbb{G} \setminus \{\mathbf{g}\}} P(G_{\mathbf{h}} = 1|X)} \quad \text{for some } \mathbf{g} \in \mathbb{G} \\
&= \frac{P(G_{\mathbf{g}} = 1|X, G_{\mathbf{g}} + (1 - D_t)(1 - G_{\mathbf{g}}) = 1) + M}{1 + M} \quad M = \frac{\sum_{\mathbf{h} \in \mathbb{G} \setminus \{\mathbf{g}\}} P(G_{\mathbf{h}} = 1|X)}{P(G_{\mathbf{g}} + (1 - D_t)(1 - G_{\mathbf{g}}) = 1|X)} \\
&< \frac{(1 - \epsilon) + M}{1 + M} \quad A3 \\
&< 1
\end{aligned}$$

### B.3 Proof of Theorem 2

In the subsample  $\mathbb{I}(\mathbf{g})$ , all units with  $\min G_i = \min \mathbf{g}$  but  $G_i \neq \mathbf{g}$  are excluded. Therefore:

- $G_{\min \mathbf{g}} = 1 \iff G_{\mathbf{g}} = 1.$
- $G_{\min \mathbf{g}} + (1 - D_t)(1 - G_{\min \mathbf{g}}) = 1 \iff (1 - D_t)(1 - G_{\mathbf{g}}) = 1.$
- $G_{\min \mathbf{g}} = 0 \iff G_{\mathbf{g}} = 0.$

We want to show that in the subsample  $\mathbb{I}(\mathbf{g})$ , each identifying moment equals the corresponding CS moment for cohort  $g = \min \mathbf{g}$ .

**Claim 8** *In the subsample  $\mathbb{I}(\mathbf{g})$ ,*

$$ATT_{unc}(\mathbf{g}, t) = ATT_{unc}^{ny, CS}(\min \mathbf{g}, t; 0),$$

where  $ATT_{unc}^{ny, CS}(g, t; \delta) = \mathbb{E}[Y_t - Y_{g-\delta-1} | G_g = 1] - \mathbb{E}[Y_t - Y_{g-\delta-1} | D_{t+\delta} = 0]$  as in CS equation 2.9.

*Proof:*

$$\begin{aligned} & ATT_{unc}^{ny, CS}(\min \mathbf{g}, t; 0) \\ &= \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} | G_{\min \mathbf{g}} = 1] - \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} | D_t = 0] && \text{CS Eq. 2.9} \\ &= \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} | G_{\min \mathbf{g}} = 1, i \in \mathbb{I}(\mathbf{g})] - \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} | D_t = 0, i \in \mathbb{I}(\mathbf{g})] && i \in \mathbb{I}(\mathbf{g}) \\ &= \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} | G_{\mathbf{g}} = 1] - \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} | D_t = 0] && \text{equivalences above} \\ &= ATT_{unc}(\mathbf{g}, t). \end{aligned}$$

Then, the inverse probability weighting version.

**Claim 9** *In the subsample  $\mathbb{I}(\mathbf{g})$ ,*

$$ATT_{ipw}(\mathbf{g}, t) = ATT_{ipw}^{ny, CS}(\min \mathbf{g}, t; 0),$$

where  $ATT_{ipw}^{ny, CS}(g, t; \delta)$  is defined as in CS equation 2.5.

*Proof:* For the propensity score  $p_{g,t}^{CS}(X) \equiv P(G_g = 1 | X, G_g + (1 - D_t)(1 - G_g) = 1)$ , we have

$$\begin{aligned}
& p_{\min \mathbf{g}, t}^{\text{CS}}(X) \\
& = P\left(G_{\min \mathbf{g}} = 1 \mid X, G_{\min \mathbf{g}} + (1 - D_t)(1 - G_{\min \mathbf{g}}) = 1, i \in \mathbb{I}(\mathbf{g})\right) \quad i \in \mathbb{I}(\mathbf{g}) \\
& = P\left(G_{\mathbf{g}} = 1 \mid X, (1 - D_t)(1 - G_{\mathbf{g}}) = 1\right) \quad \text{equivalences above} \\
& = p_{\mathbf{g}, t}(X).
\end{aligned}$$

For the inverse propensity weight  $w^{\text{comp}, \text{CS}}(\mathbf{g}, t) \equiv \frac{p_{\mathbf{g}, t}^{\text{CS}}(X)(1 - D_t)(1 - G_{\mathbf{g}})}{1 - p_{\mathbf{g}, t}^{\text{CS}}(X)}$ .

$$\begin{aligned}
w^{\text{comp}, \text{CS}}(\min \mathbf{g}, t) & = \frac{p_{\min \mathbf{g}, t}^{\text{CS}}(X)(1 - D_t)(1 - G_{\min \mathbf{g}})}{1 - p_{\min \mathbf{g}, t}^{\text{CS}}(X)} \\
& = \frac{p_{\mathbf{g}, t}(X)(1 - D_t)(1 - G_{\mathbf{g}})}{1 - p_{\mathbf{g}, t}(X)} \quad i \in \mathbb{I}(\mathbf{g}), p^{\text{CS}} = p_{\mathbf{g}} \\
& = w^{\text{comp}}(\mathbf{g}, t).
\end{aligned}$$

Now, for the IPW moment:

$$\begin{aligned}
& ATT_{ipw}^{\text{ny}, \text{CS}}(\min \mathbf{g}, t; 0) \\
& = \mathbb{E}\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{w^{\text{comp}, \text{CS}}(\mathbf{g}, t)}{\mathbb{E}[w^{\text{comp}, \text{CS}}(\mathbf{g}, t)]}\right)(Y_t - Y_{g-1}) \mid g = \min \mathbf{g}\right] \quad \text{CS Eq. 2.5} \\
& = \mathbb{E}\left[\left(\frac{G_{\mathbf{g}}}{\mathbb{E}[G_{\mathbf{g}}]} - \frac{w^{\text{comp}}(\mathbf{g}, t)}{\mathbb{E}[w^{\text{comp}}(\mathbf{g}, t)]}\right)(Y_t - Y_{\min \mathbf{g}-1})\right] \quad i \in \mathbb{I}(\mathbf{g}) \\
& = ATT_{ipw}(\mathbf{g}, t).
\end{aligned}$$

Then, the outcome regression version.

**Claim 10** In the subsample  $\mathbb{I}(\mathbf{g})$ ,

$$ATT_{or}(\mathbf{g}, t) = ATT_{or}^{\text{ny}, \text{CS}}(\min \mathbf{g}, t; 0),$$

where  $ATT_{or}^{\text{ny}, \text{CS}}$  is defined as in CS equation 2.6.

*Proof:* For the control function  $m_{\mathbf{g}, t}^{\text{CS}}(X) \equiv \mathbb{E}[Y_t - Y_{g-1} \mid X, D_t = 0, G_{\mathbf{g}} = 0]$ , we have

$$\begin{aligned}
m_{\min \mathbf{g}, t}^{\text{CS}}(X) & = \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} \mid X, D_t = 0, G_{\min \mathbf{g}} = 0, i \in \mathbb{I}(\mathbf{g})] \\
& = \mathbb{E}[Y_t - Y_{\min \mathbf{g}-1} \mid X, D_t = 0, G_{\mathbf{g}} = 0] \quad \text{equivalences above} \\
& = m_{\mathbf{g}, t}^{\text{ny}}(X).
\end{aligned}$$

Then for the identifying moment:

$$\begin{aligned}
& ATT_{or}^{ny,CS}(\min \mathbf{g}, t; 0) \\
&= \mathbb{E} \left[ \frac{G_g}{\mathbb{E}[G_g]} (Y_t - Y_{g-1} - m_{g,t}^{ny}(X)) \middle| g = \min \mathbf{g} \right] && \text{CS Eq. 2.6} \\
&= \mathbb{E} \left[ \frac{G_{\mathbf{g}}}{\mathbb{E}[G_{\mathbf{g}}]} (Y_t - Y_{\min \mathbf{g}-1} - m_{\mathbf{g},t}^{ny}(X)) \right] && i \in \mathbb{I}(\mathbf{g}) \\
&= ATT_{or}(\mathbf{g}, t).
\end{aligned}$$

Finally, for the doubly-robust:

**Claim 11** In the subsample  $\mathbb{I}(\mathbf{g})$ ,

$$ATT_{dr}(\mathbf{g}, t) = ATT_{dr}^{ny,CS}(\min \mathbf{g}, t; 0),$$

where  $ATT_{dr}^{ny,CS}$  is as defined in CS equation 2.7.

*Proof:*

$$\begin{aligned}
& ATT_{dr}^{ny,CS}(\min \mathbf{g}, t; 0) \\
&= \mathbb{E} \left[ \left( \frac{G_g}{\mathbb{E}[G_g]} - \frac{w^{comp,CS}(g, t)}{\mathbb{E}[w^{comp,CS}(g, t)]} \right) (Y_t - Y_{g-1} - m_{g,t}^{ny}(X)) \middle| g = \min \mathbf{g} \right] && \text{CS Eq. 2.7} \\
&= \mathbb{E} \left[ \left( \frac{G_{\mathbf{g}}}{\mathbb{E}[G_{\mathbf{g}}]} - \frac{w^{comp}(\mathbf{g}, t)}{\mathbb{E}[w^{comp}(\mathbf{g}, t)]} \right) (Y_t - Y_{\min \mathbf{g}-1} - m_{\mathbf{g},t}^{ny}(X)) \right] && i \in \mathbb{I}(\mathbf{g}) \\
&= ATT_{dr}(\mathbf{g}, t).
\end{aligned}$$

With the equivalence of moments established, Theorem 1 in CS provides identification. For  $ATT_x \in \{ATT_{unc}, ATT_{ipw}, ATT_{or}, ATT_{dr}\}$ :

$$\begin{aligned}
ATT_x(\mathbf{g}, t) &= ATT_x^{ny,CS}(\min \mathbf{g}, t; 0) && \text{Claims 1--4} \\
&= ATT(\min \mathbf{g}, t) && \text{CS Theorem 1} \\
&= \mathbb{E}[Y_t^{cs}(\min \mathbf{g}) - Y_t^{cs}(0) \mid G_{\min \mathbf{g}} = 1, i \in \mathbb{I}(\mathbf{g})] \\
&= \mathbb{E}[Y_t(\mathbf{g}) - Y_t(\mathbf{0}) \mid G_{\mathbf{g}} = 1] && Y_t^{cs}(\min \mathbf{g}) = Y_t(\mathbf{g}), Y_t^{cs}(0) = Y_t(\mathbf{0}) \\
&= ATT(\mathbf{g}, t).
\end{aligned}$$

In the full sample  $\mathbb{I}$ ,  $ATT(\mathbf{g}, t)|_{i \in \mathbb{I}(\mathbf{g})} = ATT(\mathbf{g}, t)$  since  $G_{\mathbf{g}} = 1 \implies i \in \mathbb{I}(\mathbf{g})$ .



## B.4 Proof of Theorem 3

Let  $g' = \min_{d \in \mathbf{M} \setminus \{1\}} g^d$ .

**Case 1:**  $t < g'$ .

$$\begin{aligned}
 ATT_{id}^{dd}(\mathbf{g}, t) &= ATT_{id}(\mathbf{g}, t) && \text{Eq. 14, } t < g' \\
 &= \text{ATT}(\mathbf{g}, t) && \text{Theorem 2} \\
 &= \text{ATT}^1(\mathbf{g}, t) && \text{Eq. 9, } t < g' \leq g^d \ \forall d \neq 1 \\
 &= ATT^1(\mathbf{g}, t).
 \end{aligned}$$

**Case 2:**  $t \geq g', g^1 < g'$  (**imputation**).

Let  $b = g' - 1$ . Since  $g^1 < g'$ , event 1 has occurred by  $b$  while no  $d \neq 1$  has ( $g^d \geq g' > b$  for all  $d \neq 1$ ). By Equation 9,

$$\text{ATT}(\mathbf{g}, b) = \text{ATT}^1(\mathbf{g}, b).$$

For any  $\mathbf{h} \in C^{imp}(\mathbf{g}, t)$  (so  $h^1 = g^1$  and  $h^d > t \geq g' > b$  for all  $d \neq 1$ ), no confounding event has occurred at  $b$  or  $t$ , so:

$$\text{ATT}(\mathbf{h}, b) = \text{ATT}^1(\mathbf{h}, b), \quad \text{ATT}(\mathbf{h}, t) = \text{ATT}^1(\mathbf{h}, t).$$

By Assumption 7 (same  $h^1 = g^1$ ),

$$\text{ATT}^1(\mathbf{g}, t) - \text{ATT}^1(\mathbf{g}, b) = \text{ATT}^1(\mathbf{h}, t) - \text{ATT}^1(\mathbf{h}, b).$$

Therefore:

$$\begin{aligned}
 ATT_{id}^{dd}(\mathbf{g}, t) &= ATT_{id}^{imp}(\mathbf{g}, t) && \text{Eq. 14, } g^1 < g' \\
 &= \text{ATT}(\mathbf{g}, b) + \mathbb{E}[\text{ATT}(\mathbf{h}, t) - \text{ATT}(\mathbf{h}, b) \mid \mathbf{h} \in C^{imp}(\mathbf{g}, t)] \\
 &= \text{ATT}^1(\mathbf{g}, b) + \mathbb{E}[\text{ATT}^1(\mathbf{h}, t) - \text{ATT}^1(\mathbf{h}, b) \mid \mathbf{h} \in C^{imp}(\mathbf{g}, t)] \\
 &= \text{ATT}^1(\mathbf{g}, b) + \text{ATT}^1(\mathbf{g}, t) - \text{ATT}^1(\mathbf{g}, b) && \text{Ass. 7} \\
 &= \text{ATT}^1(\mathbf{g}, t).
 \end{aligned}$$

**Case 3:**  $t \geq g', g^1 > g'$  (**DiD**).

Let  $b = g^1 - 1$ . For any  $\mathbf{h} \in C^{did}(\mathbf{g}, t)$  ( $h^1 > t$  and  $h^d = g^d$  for all  $d \neq 1$  with  $g^d \leq t$ ), neither  $\mathbf{g}$  nor  $\mathbf{h}$  has had event 1 at  $b$ , and  $h^d = g^d$  for all  $d \neq 1$  with  $g^d \leq b < t$ . By Equation 9:

$$\text{ATT}(\mathbf{g}, b) = \sum_{d \neq 1: g^d \leq b} \text{ATT}^d(\mathbf{g}, b), \quad \text{ATT}(\mathbf{h}, b) = \sum_{d \neq 1: g^d \leq b} \text{ATT}^d(\mathbf{h}, b).$$

Since  $h^d = g^d$  for each such  $d$ , Assumption 7 gives  $\text{ATT}^d(\mathbf{g}, b) = \text{ATT}^d(\mathbf{h}, b)$ , hence:

$$\text{ATT}(\mathbf{g}, b) - \text{ATT}(\mathbf{h}, b) = 0.$$

At  $t$ , since  $h^1 > t$ :

$$\text{ATT}(\mathbf{g}, t) = \text{ATT}^1(\mathbf{g}, t) + \sum_{d \neq 1: g^d \leq t} \text{ATT}^d(\mathbf{g}, t), \quad \text{ATT}(\mathbf{h}, t) = \sum_{d \neq 1: g^d \leq t} \text{ATT}^d(\mathbf{h}, t).$$

By Assumption 7,  $\text{ATT}^d(\mathbf{g}, t) = \text{ATT}^d(\mathbf{h}, t)$  for each  $d \neq 1$  with  $g^d \leq t$ . Therefore:

$$\begin{aligned} \text{ATT}_{id}^{dd}(\mathbf{g}, t) &= \text{ATT}_{id}^{did}(\mathbf{g}, t) && \text{Eq. 14, g} \\ &= \text{ATT}(\mathbf{g}, t) - \underbrace{\left( \text{ATT}(\mathbf{g}, b) - \mathbb{E}[\text{ATT}(\mathbf{h}, b) \mid \mathbf{h} \in C^{did}(\mathbf{g}, t)] \right)}_{=0} - \mathbb{E}[\text{ATT}(\mathbf{h}, t) \mid \mathbf{h} \in C^{did}(\mathbf{g}, t)] \\ &= \text{ATT}^1(\mathbf{g}, t) + \sum_{d \neq 1: g^d \leq t} \text{ATT}^d(\mathbf{g}, t) - \sum_{d \neq 1: g^d \leq t} \text{ATT}^d(\mathbf{g}, t) && \text{Ass. 7} \\ &= \text{ATT}^1(\mathbf{g}, t). \end{aligned}$$